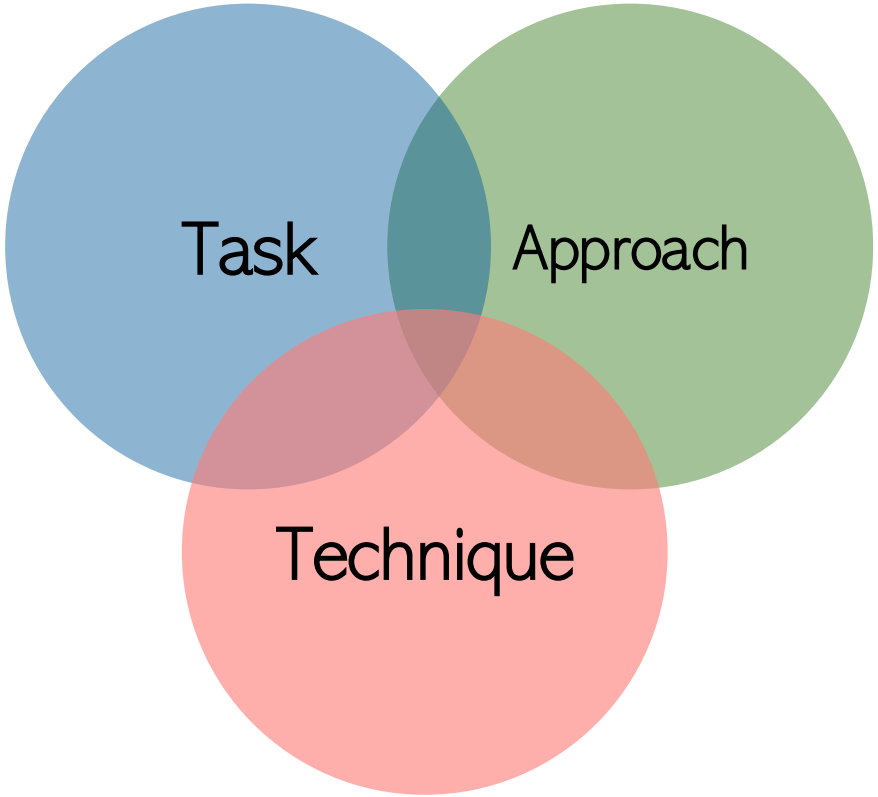


Free solo climbing (or free soloing) is rock climbing alone without any ropes, harnesses, or safety protection
https://films.nationalgeographic.com/free-solo

2

Outline: W07C: New Technique: Logistic Regression (LR)

Classification
Regression
Clustering
Ranking (Recommendation)
Vector Representation and Dimensionality Reduction
Language Modeling



Full/Self/Semi-**Supervised Learning**
Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, **Logistic Regression**
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

Outline: W07C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: Logistic Regression → Neural Networks

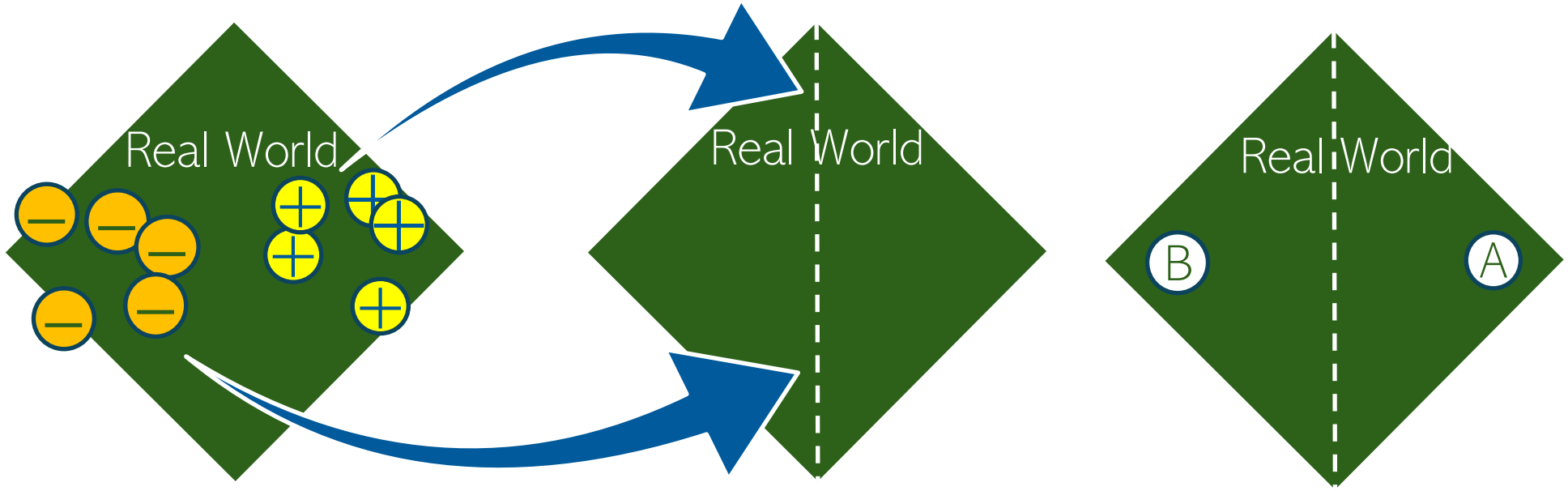
IIII) Evaluation

Metrics (Few New Metrics), Methodology

4

A Technique based on a Global Discriminator

Summarizing the Entire World into a Discrimination Boundary



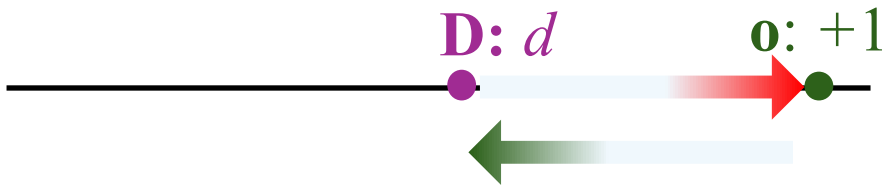
f(A) = Far Right = Very Pos
f(B) = Far Left = Very Neg

5

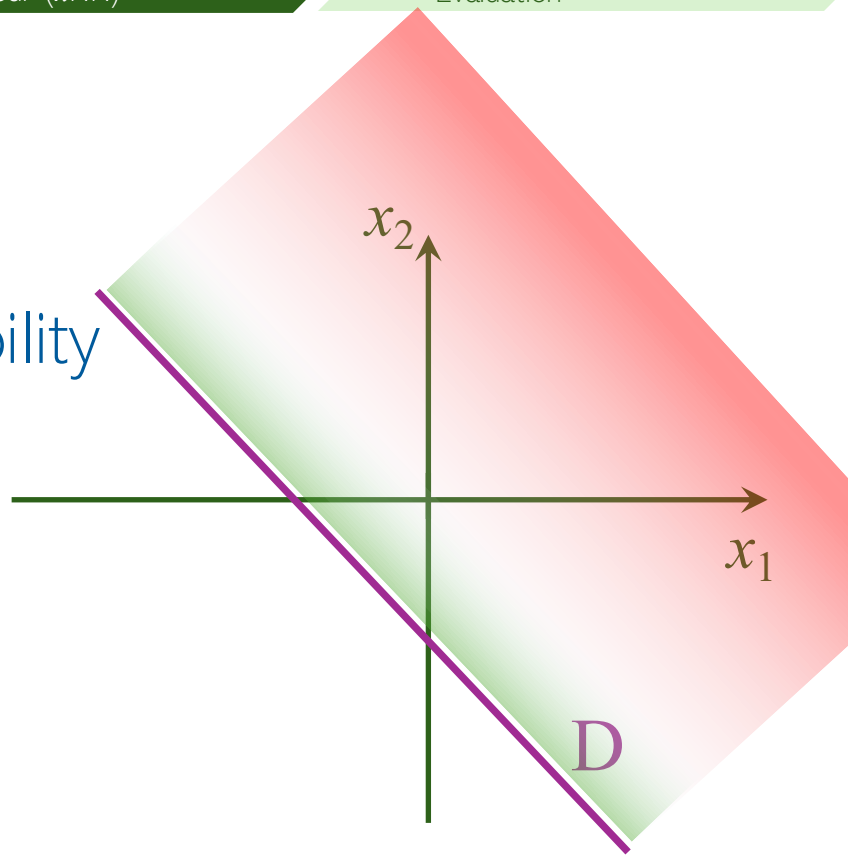
Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

$$
\begin{aligned}
f(\mathbf{o}) &= d(\mathbf{o}, \mathrm{D}) \\
&= logistic\ curve(o_i - d) \\
&= logistic\ regression(o_i - d) \\
&= sigmoid(o_i - d) \\
&= \sigma\,(o_i - d) \\
&= \begin{cases} \text{The more toward } +1\text{, stronger probability for being positive} \\ \underbrace{1 - \text{probability for being positive}}_{\text{Being negative}} = 1 - sigmoid(o_i - d). \end{cases}
\end{aligned}
$$



Discriminator: Euclidian Space



2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

$$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$$

$$\approx sigmoid(\begin{bmatrix} x_1 & x_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix})$$

$$\approx \sigma(\begin{bmatrix} x_1 & x_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix})$$

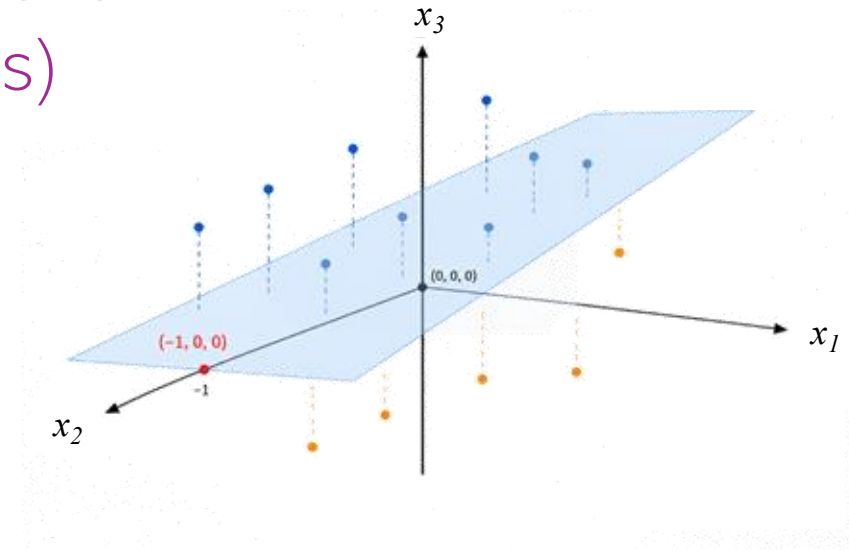
$$\approx \begin{cases} \text{The more toward } +1\text{, stronger } \textbf{probability} \text{ for being positive} \\ \underbrace{1 - \text{probability for being positive}}_{\text{Being negative}} = 1 - sigmoid(\begin{bmatrix} x_1 & x_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}) \end{cases}$$

7 Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^3 \rightarrow \mathbb{R}^2$

The universe can be summarized (learned) into a plane:

- 4 parameters (coefficients, weights, memory cells)



$\rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

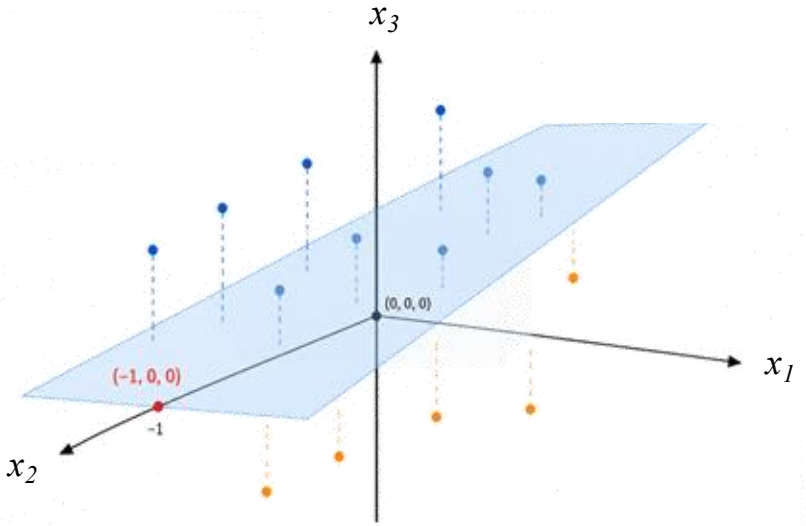
$$f(\mathbf{o}) = d(\mathbf{o}, \mathbf{D}) \approx \sigma([x_1 \quad x_2 \quad x_3 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix})$$

$$\approx \begin{cases} \text{The more toward } +1\text{, stronger } \textbf{probability} \text{ for being positive} \\ \underbrace{1 - \text{probability for being positive}}_{\text{Being negative}} = 1 - sigmoid(\ldots) \end{cases}$$

Discriminator: Euclidian Space

What is, or who knows, the true discriminator **D**?

- Point, what is that point $d=w_1$?
- Line, what is that line w_1, w_2, w_3?
- Plane, what is that plane w_1, w_2, w_3, w_4?
- Hyperplane, what is that hyperplane w_1, w_2, …, w_{n+1}?



We find it through an iterative process:

1. Initial random guess
2. Evaluate on the train set
3. Tune the discriminator's parameters toward *almost* overfitting
4. Back to (2)





Outline: W08A

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: Logistic Regression

Learning the Discriminator from Training Data (from Experience)

Mountain Descending

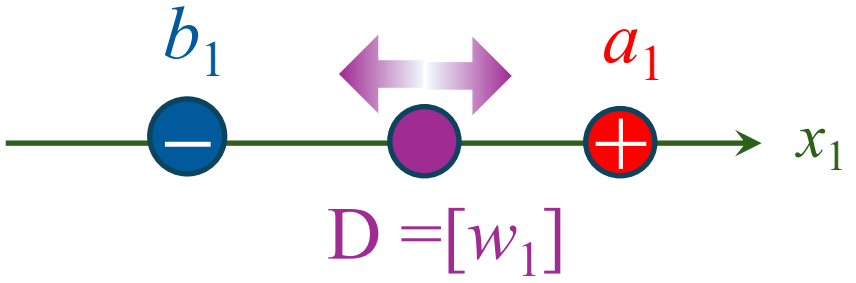
IIII) Evaluation

Metrics (Few New Metrics), Methodology

10

Learning the Discriminator

Both positive and negative observations



A. Error $= [f^+(\boldsymbol{o}^+) - \text{'Positive'}] + [f^+(\boldsymbol{o}^-) - \text{'Negative'}]$

$= [1.0 - f^+(\boldsymbol{o}^+)] + [f^+(\boldsymbol{o}^-) - 0.0]$

$= [1.0 - \sigma(a_1 - w_1)] + [\sigma(b_1 - w_1)]$

$= e(w_1) \geq 0$

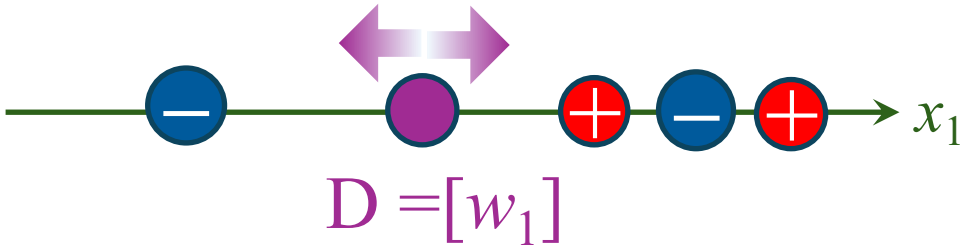
B. Modify w_1 such that $e(w_1) \rightarrow 0.0$

C. Go back to A

Learning the Discriminator → An Optimization Problem
Data Mining Task → Mathematic Task

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$



No solution may exist (no real value of w_1 satisfies the equation exactly.)

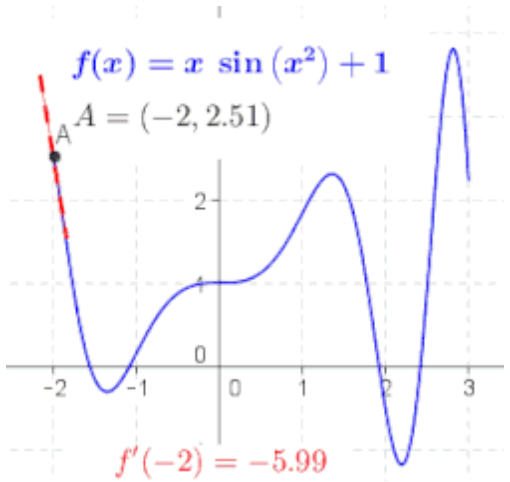
No closed-form solution may exist (a solution exists, but difficult to obtain.)

An Optimization Problem

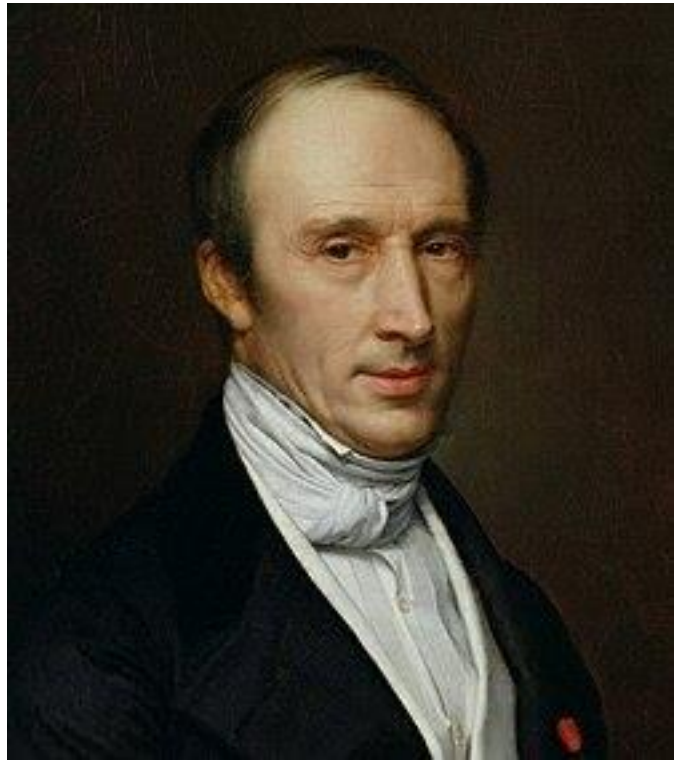


Pierre de Fermat (1700 – 1665)
French magistrate, polymath, and above all, mathematician also a poet
He is notably credited with Fermat's principle in optics.
He is especially known for having stated Fermat's Last Theorem, the proof was only established more than 300 years later.

Fermat noticed that at the very peak (maximum) or very trough /traaf/ (minimum) of a curve, the curve briefly flattens out. If you shift your position by an infinitely small amount, the output value barely changes



Baron Augustin-Louis Cauchy (1789 – 1857)
French mathematician, engineer, and physicist.
Proved the key theorems of calculus more than any other mathematician.
Approximately **800 research articles and 5 complete textbooks**.



While Fermat mastered finding minimums for single-variable equations, optimizing complex functions with multiple variables required the gradient. Cauchy invented the **Gradient Descent algorithm** to solve complex astronomical and algebraic equations, recognizing that moving in the **opposite direction of the gradient vector** provides the **steepest route** to a minimum point.



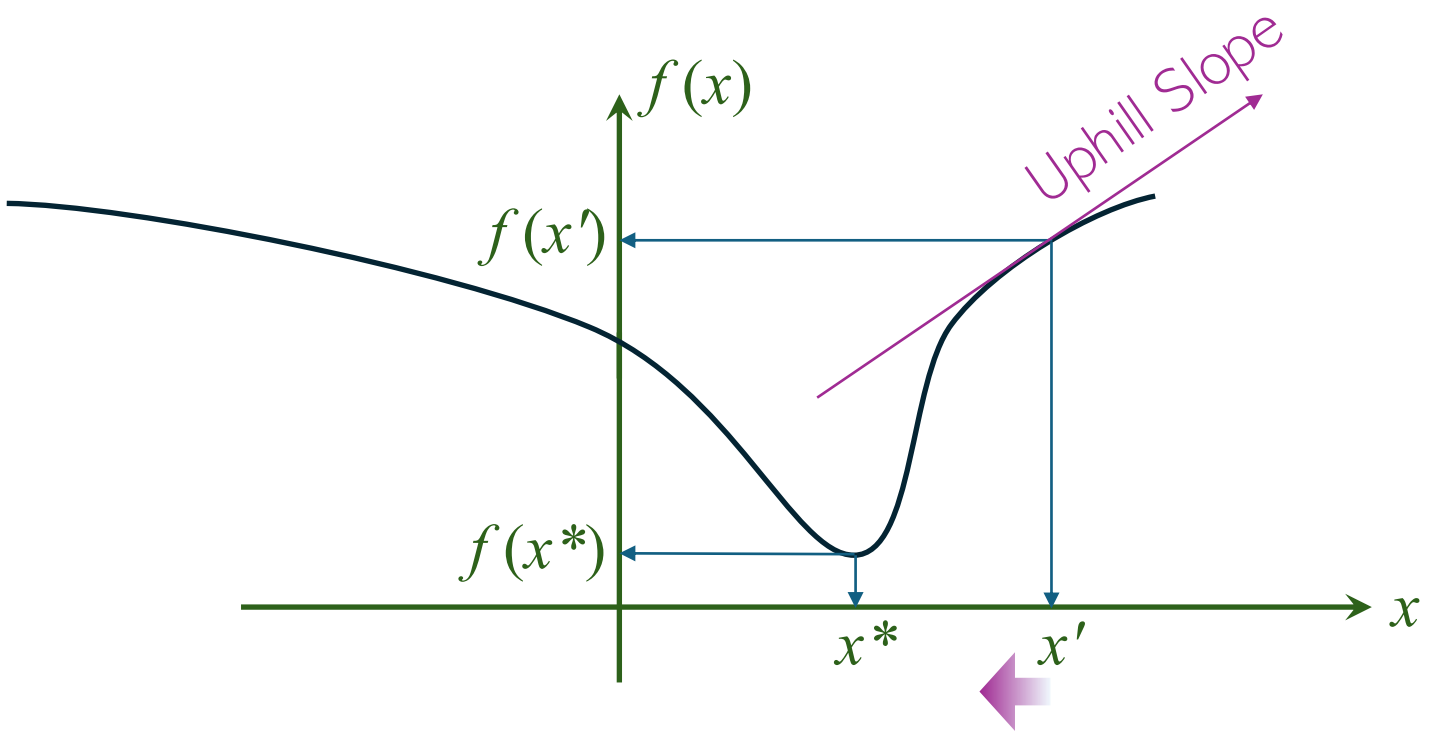
Mathematic Task: Finding the Minimum of a Function

Differentiation, Derivative

Where $f(x)$ become minimum value?

At what $x=x^*$, $f(x=x^*)$ become minimum value?

If at x', how to modify $x' \rightarrow x^*$ such that $f(x)$ get closer to minimum?



$$= lim_{\Delta x \rightarrow 0} \frac{rise}{run} = lim_{\Delta x \rightarrow 0} \frac{f(x'+\Delta x) - f(x')}{x' - (x'+\Delta x)}$$

$$= lim_{\Delta x \rightarrow 0} \frac{f(x'+\Delta x) - f(x')}{\Delta x}$$

$$= \frac{d\,f(x')}{d\,x} > 0$$

One step down:

$$x' \rightarrow x' - \frac{d\,f(x')}{d\,x}$$



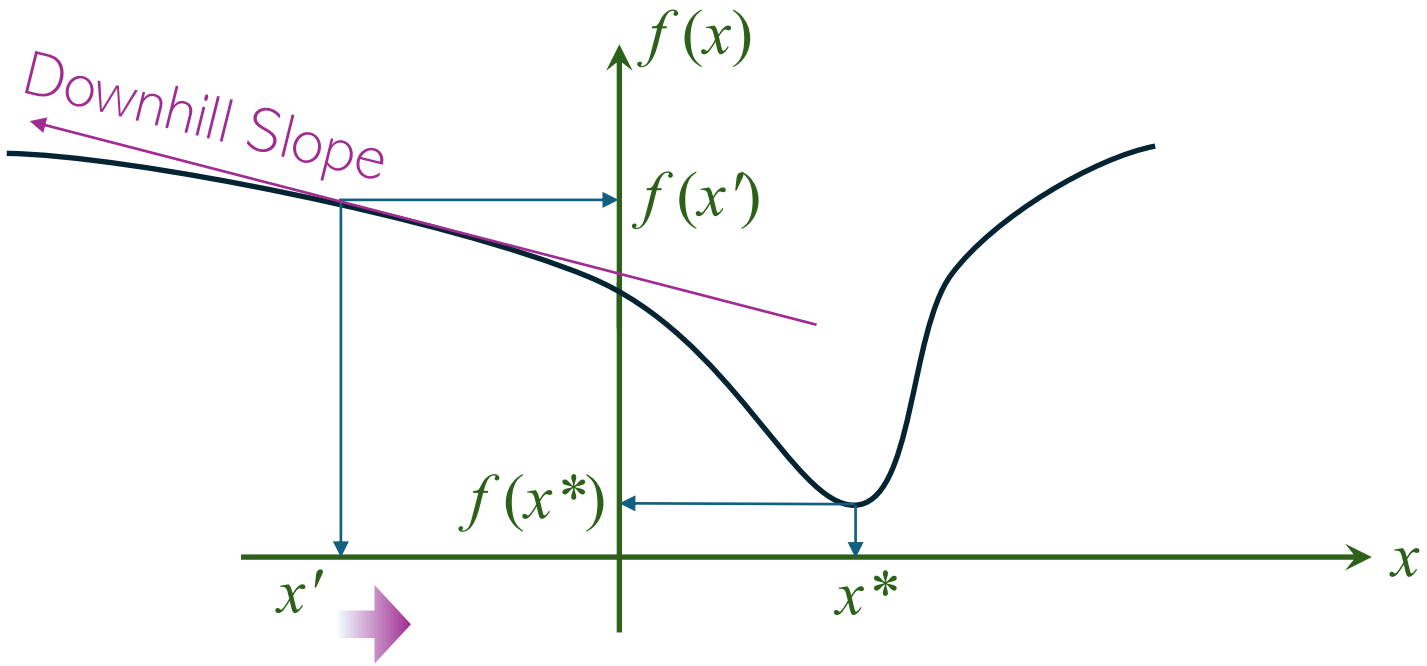
Mathematic Task: Finding the Minimum of a Function

Differentiation, Derivative

Where $f(x)$ become minimum value?

At what $x=x^*$, $f(x=x^*)$ become minimum value?

If at x', how to modify $x' \rightarrow x^*$ such that $f(x)$ get closer to minimum?



$$= lim_{\Delta x \to 0} \frac{rise}{run} = lim_{\Delta x \to 0} \frac{f(x'+\Delta x) - f(x')}{x' - (x'+\Delta x)}$$

$$= lim_{\Delta x \to 0} \frac{f(x'+\Delta x) - f(x')}{\Delta x}$$

$$= \frac{d f(x')}{d x} < 0$$

One step down:

$$x' \rightarrow x' - \frac{d f(x')}{d x}$$

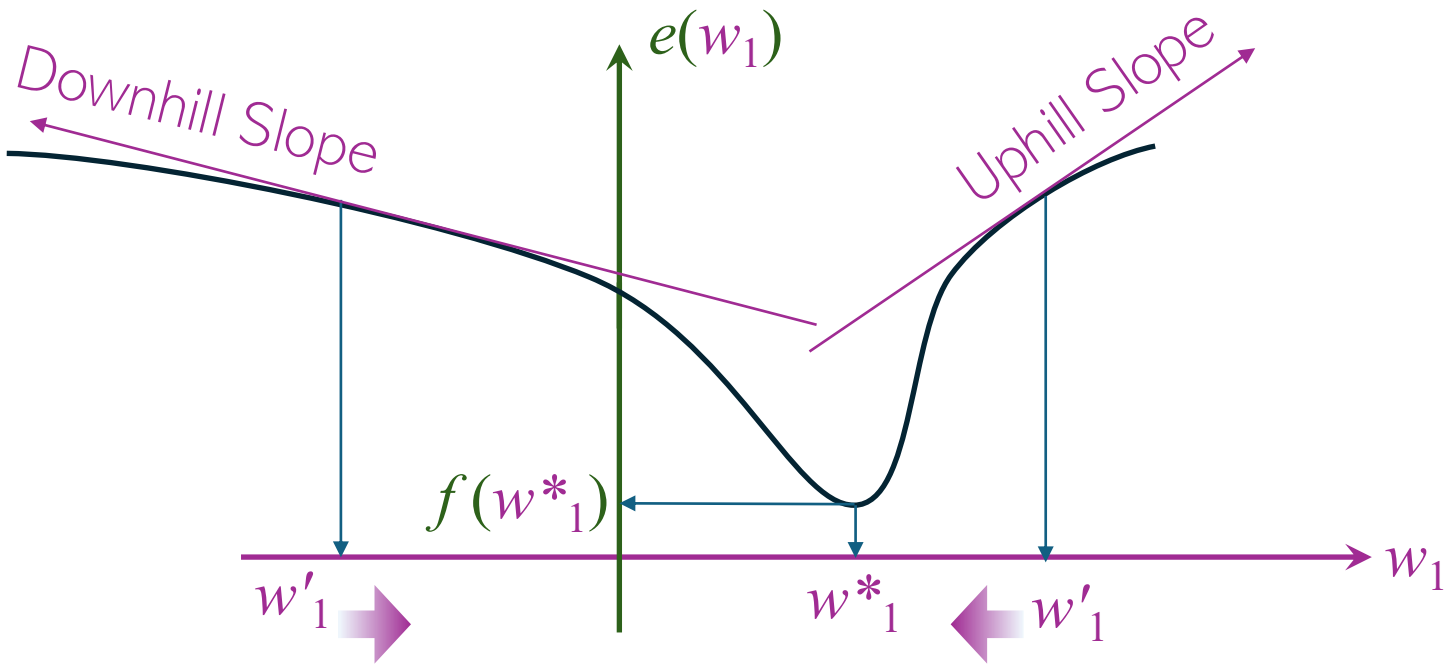
Back to Data Mining Task: Minimizing LR Error

Differentiation, Derivative

Where $e(w_1)$ become minimum value?

At what $w_1 = w^*_1$, $e(w_1 = w^*_1)$ become minimum value?

If at w'_1, how to modify $w'_1 \rightarrow w^*_1$ such that $e(w_1)$ get closer to minimum?

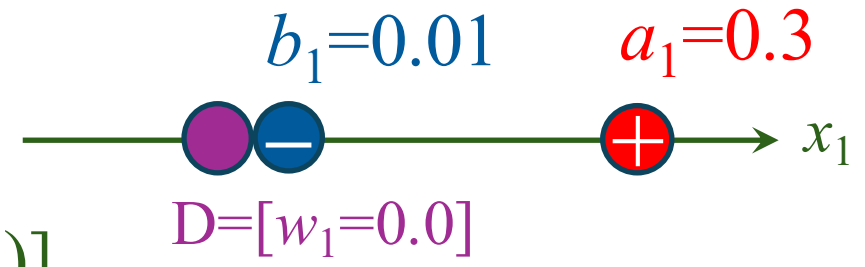


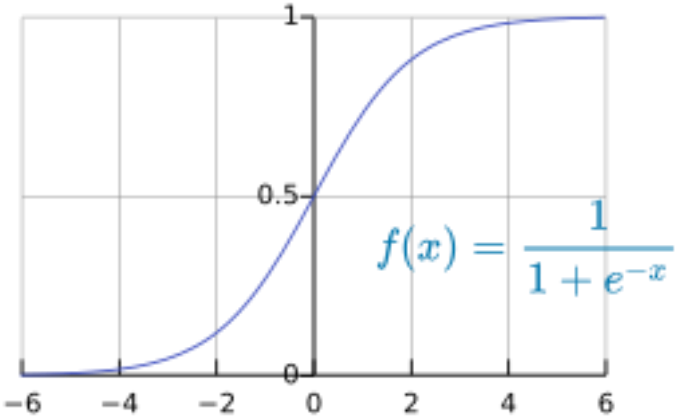
One step down:

$$w'_1 \rightarrow w'_1 - \frac{d\ e(w_1)}{d\ w_1}$$

Back to Data Mining Task: Minimizing LR Error

$$
\begin{aligned}
e\,(w_1 = 0.0) &= [1.0 - f^{+}(\boldsymbol{o}^{+})] + [f^{+}(\boldsymbol{o}^{-}) - 0.0] \\
&= [1.0 - f^{+}(a_1)] + [f^{+}(b_1)] \\
&= [1.0 - \sigma\,(0.3 - w_1)] + [\sigma\,(0.01 - w_1)] \\
&= [1.0 - \sigma\,(0.3 - 0.0)] + [\sigma\,(0.01 - 0.0)] \\
&= [1.0 - \sigma\,(0.3)] + [\sigma\,(0.01)] \\
&= [1.0 - 0.57] + [0.50] \\
&= [0.42] + [0.50] \\
&= +\,0.92
\end{aligned}
$$

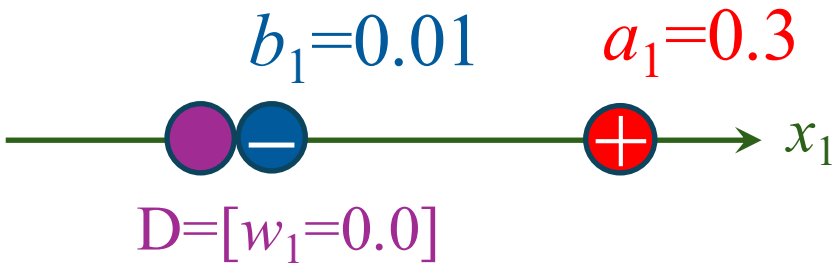




https://www.desmos.com/calculator

Back to Data Mining Task: Minimizing LR Error

$$
\begin{aligned}
e(w_1) &= [1.0 - f^+(\boldsymbol{o}^+)] + [f^+(\boldsymbol{o}^-) - 0.0] \\
&= [1.0 - f^+(a_1)] + [f^+(b_1)] \\
&= [1.0 - \sigma(0.3 - w_1)] + [\sigma(0.01 - w_1)]
\end{aligned}
$$



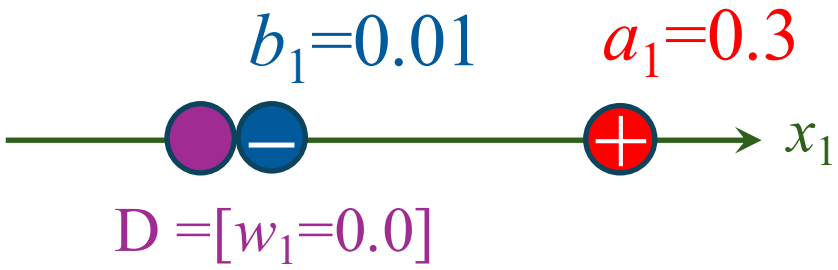
$$
\frac{d\, e(w_1)}{d\, w_1} = \frac{d\, [1.0 - \sigma(0.3 - w_1)] + [\sigma(0.01 - w_1)]}{d\, w_1} = \frac{d\ -\sigma(0.3 - w_1) + \sigma(0.01 - w_1)}{d\, w_1}
$$

$$
= \sigma(0.3 - w_1)[1 - \sigma(0.3 - w_1)] - \sigma(0.01 - w_1)[1 - \sigma(0.01 - w_1)]
$$

$$
\frac{d}{dx}\sigma(g(x)) = -\frac{1}{(1+e^{-g(x)})^2}(-e^{-g(x)}g'(x)) = \frac{e^{-g(x)}g'(x)}{(1+e^{-g(x)})^2} = \sigma(g(x))(1-\sigma(g(x)))g'(x)
$$

Back to Data Mining Task: Minimizing LR Error

One step down:



$$w'_1 \rightarrow w'_1 - \frac{d\ e(w_1)}{d\ w_1}$$

$$\rightarrow w'_1 - [\sigma(0.3 - w_1)[1 - \sigma(0.3 - w_1)] - \sigma(0.01 - w_1)[1 - \sigma(0.01 - w_1)]]$$

$$\rightarrow 0.0 - [\sigma(0.3 - 0.0)[1 - \sigma(0.3 - 0.0)] - \sigma(0.01 - 0.0)[1 - \sigma(0.01 - 0.0)]]$$

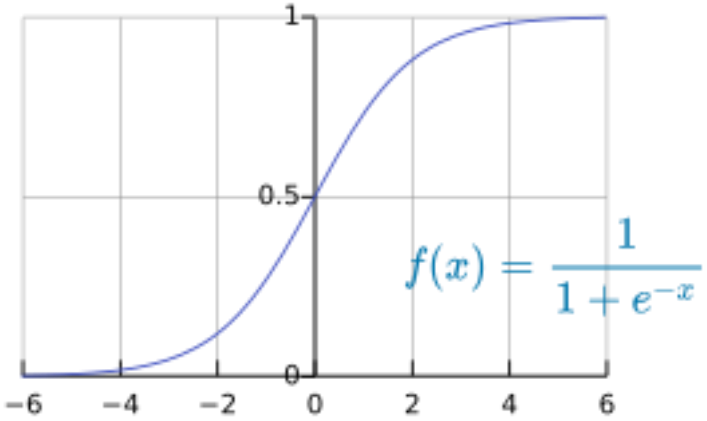
$$\rightarrow 0.0 - [\sigma(0.3)[1 - \sigma(0.3)] - \sigma(0.01)[1 - \sigma(0.01)]]$$

$$\rightarrow 0.0 - [0.57[1 - 0.57] - 0.50[1 - 0.50]]$$

$$\rightarrow 0.0 - [0.24 - 0.25]$$

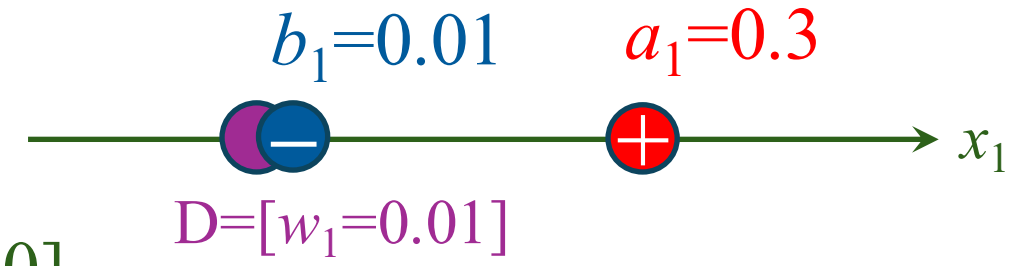
$$\rightarrow 0.0 - [-0.01]$$

$$\rightarrow 0.0 + 0.01$$

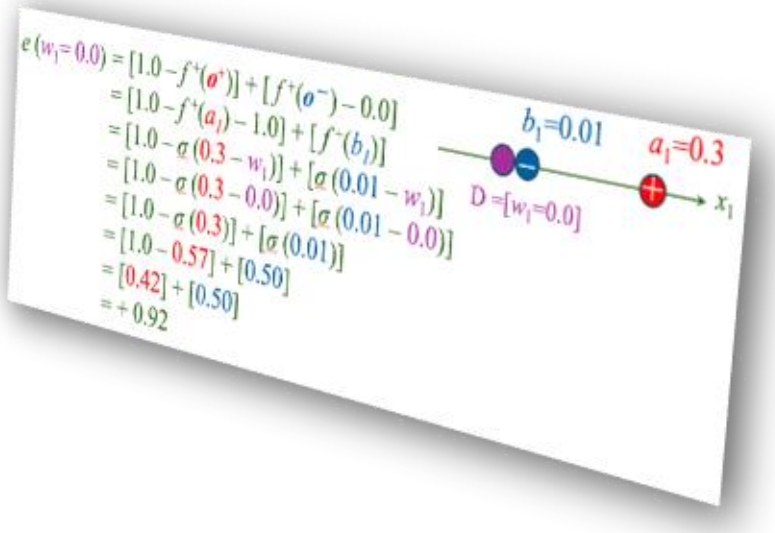


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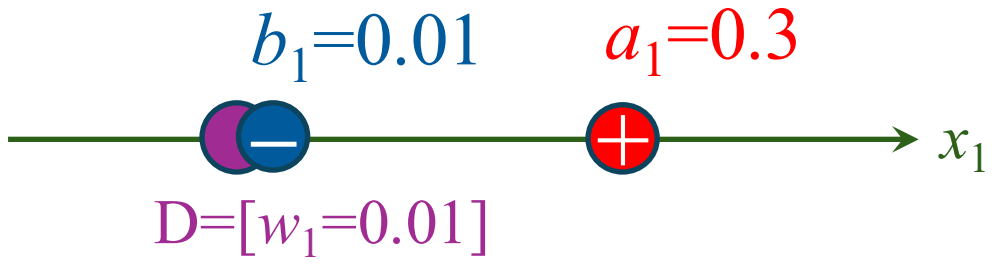
Back to Data Mining Task: Minimizing LR Error



$$
\begin{aligned}
e\,(w_1 = 0.01) &= [1.0 - f^{+}(\boldsymbol{o}^{+})] + [f^{+}(\boldsymbol{o}^{-}) - 0.0] \\
&= [1.0 - f^{+}(a_1)] + [f^{+}(b_1)] \\
&= [1.0 - \sigma\,(0.3 - w_1)] + [\sigma\,(0.01 - w_1)] \\
&= [1.0 - \sigma\,(0.3 - 0.01)] + [\sigma\,(0.01 - 0.01)\,] \\
&= [1.0 - \sigma\,(0.29)] + [\sigma\,(0.0)] \\
&= [1.0 - 0.57] + [0.50] \\
&= [0.43] + [0.50] \\
&= +\,0.93
\end{aligned}
$$



Back to Data Mining Task: Minimizing LR Error



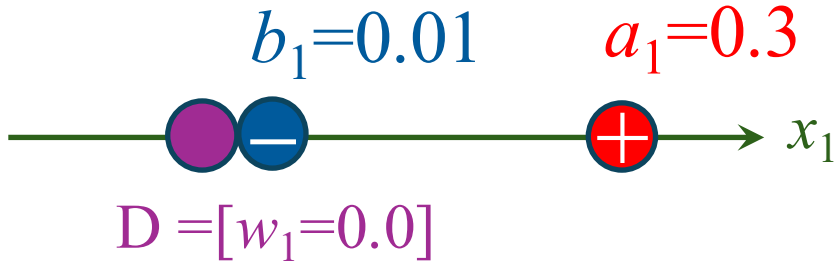
$$e(w^{(1)}{}_1 = 0.01) = +0.93 \; > \; e(w^{(0)}{}_1 = 0.00) = +0.92$$

- We were already at minimum, or
- We took a large backward step,
  → We passed the point where minimum happens.

- Or many other reasons (we may discuss them in future …)

Learning the Discriminator: Dynamic Steps: Learning Rate

$$
\begin{aligned}
e\,(w_1) &= [1.0 - f^{+}(\boldsymbol{o}^{+})] + [f^{+}(\boldsymbol{o}^{-}) - 0.0] \\
&= [1.0 - f^{+}(a_1)] + [f^{+}(b_1)] \\
&= [1.0 - \sigma\,(0.3 - w_1)] + [\sigma\,(0.01 - w_1)]
\end{aligned}
$$



When **error is high** → far from minimum error → **larger steps** …

When **error is low** → getting close → **smaller steps** …

$$w^{(1)}{}_1 \rightarrow w^{(0)}{}_1 - [\eta{=}1.0] \times \frac{d\ e(w^{(0)}{}_1 : (a_1, b_1))}{d\ w_1}$$

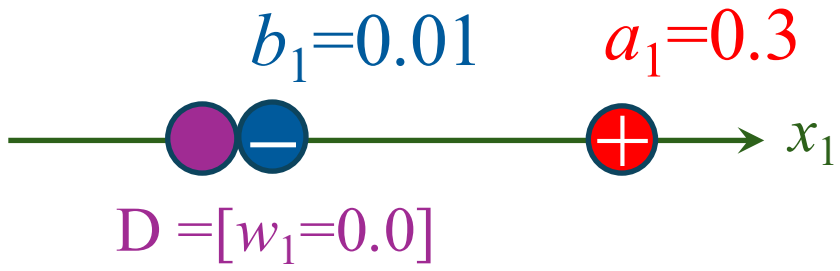
$$w^{(2)}{}_1 \rightarrow w^{(1)}{}_1 - [\eta{=}0.5] \times \frac{d\ e(w^{(1)}{}_1 : (a_1, b_1))}{d\ w_1}$$

$$w^{(3)}{}_1 \rightarrow w^{(2)}{}_1 - [\eta{=}0.1] \times \frac{d\ e(w^{(2)}{}_1 : (a_1, b_1))}{d\ w_1}$$

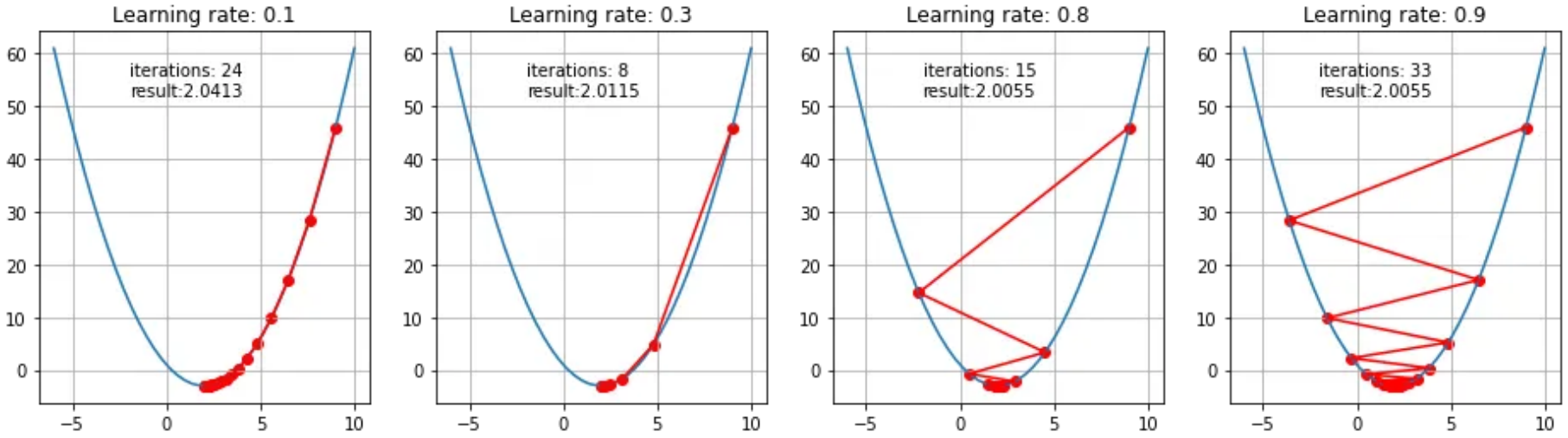
eta

Learning the Discriminator: Dynamic Steps: Learning Rate

$$e(w_1) = [1.0 - f^+(o^+)] + [f^+(o^-) - 0.0]$$
$$= [1.0 - f^+(a_1)] + [f^+(b_1)]$$
$$= [1.0 - \sigma(0.3 - w_1)] + [\sigma(0.01 - w_1)]$$



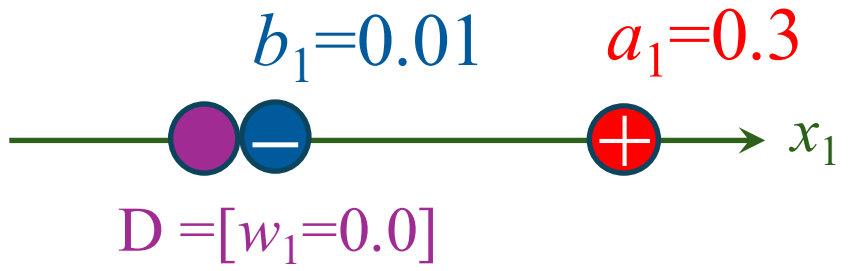
$$w^{(i+1)}{}_1 \rightarrow w^{(i)}{}_1 - \eta \times \frac{d\, e(w^{(i)}{}_1 : (a_1, b_1))}{d\, w_1}$$



https://towardsdatascience.com/gradient-descent-algorithm-a-deep-dive-cf04e8115f21

Learning the Discriminator: Early Stopping w/ Patience

$$e(w_1) = [1.0 - f^+(\boldsymbol{o}^+)] + [f^+(\boldsymbol{o}^-) - 0.0]$$
$$= [1.0 - f^+(a_1)] + [f^+(b_1)]$$
$$= [1.0 - \sigma(0.3 - w_1)] + [\sigma(0.01 - w_1)]$$



After certain number of steps (patience: e.g., 3),

→ No (substantial) drop to error or increasing error → Underfitting

→ Dropping errors but then plateau and then increasing

→ Stop Training

→ $w^{(0)}{}_1$ → $w^{(1)}{}_1$ → $w^{(2)}{}_1$ → $w^{(3)}{}_1$ → $w^{(4)}{}_1$

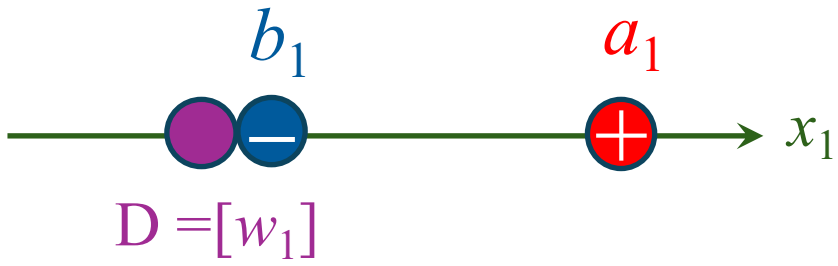
→ e($w^{(0)}{}_1$)=0.99 → e($w^{(1)}{}_1$)=0.85 → e($w^{(2)}{}_1$)=0.73 → e($w^{(3)}{}_1$)=0.72 → e($w^{(4)}{}_1$)=0.75 → Stop

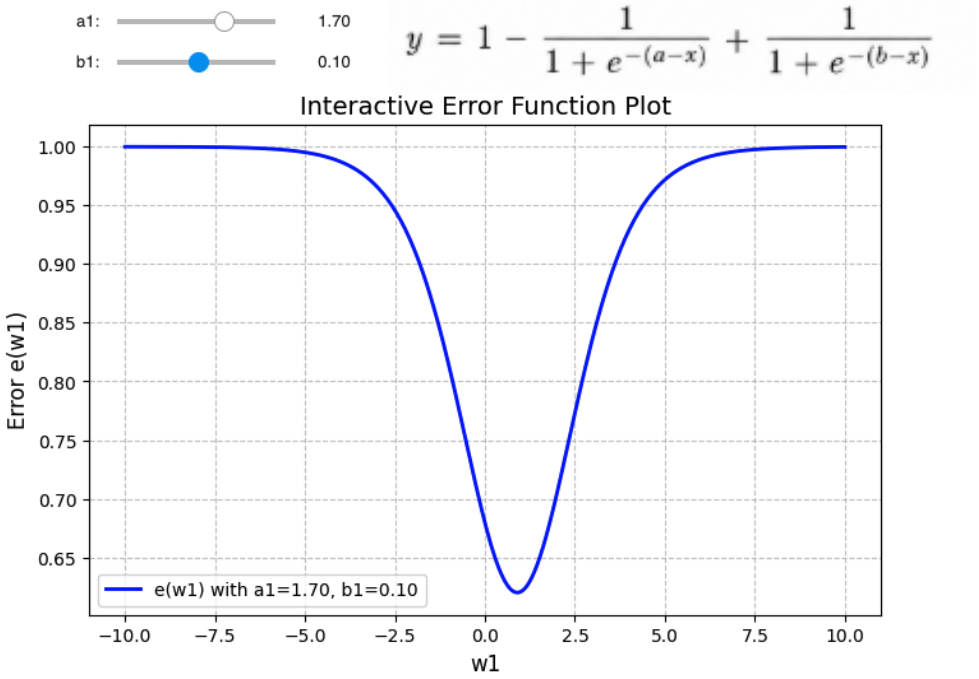
26

Learning the Discriminator: Error Function is Convex

https://chatgpt.com/s/w_6ae2afb0155881 91b86efb2e1b015c3e

$$e(w_1) = [1.0 - f^+(\boldsymbol{o}^+)] + [f^+(\boldsymbol{o}^-) - 0.0]$$
$$= [1.0 - f^+(a_1)] + [f^+(b_1)]$$
$$= [1.0 - \sigma(a_1 - w_1)] + [\sigma(b_1 - w_1)]$$





Gemini

[3] ✓ 0s

```
from ipywidgets import interact, FloatSlider

def interactive_plot(a1, b1):
    w1_vals = np.linspace(-10, 10, 500)
    error_vals = error_function(w1_vals, a1, b1)

    fig, ax = plt.subplots(figsize=(8, 5))
    ax.plot(w1_vals, error_vals, label=f'e(w1) with a1={a1:.2f}, b1={b1:.2f}',
    ax.set_title('Interactive Error Function Plot', fontsize=14)
    ax.set_xlabel('w1', fontsize=12)
    ax.set_ylabel('Error e(w1)', fontsize=12)
    ax.grid(True, linestyle='--', alpha=0.7)
    ax.legend()
    plt.show()

# Create interactive sliders for a1 and b1
interact(
    interactive_plot,
    a1=FloatSlider(value=-2.0, min=-5.0, max=5.0, step=0.1, description='a1:')
    b1=FloatSlider(value=2.0, min=-5.0, max=5.0, step=0.1, description='b1:')
);
```



Lead climbing is a technique in rock climbing where two climbers work together to ascend a climbing route.

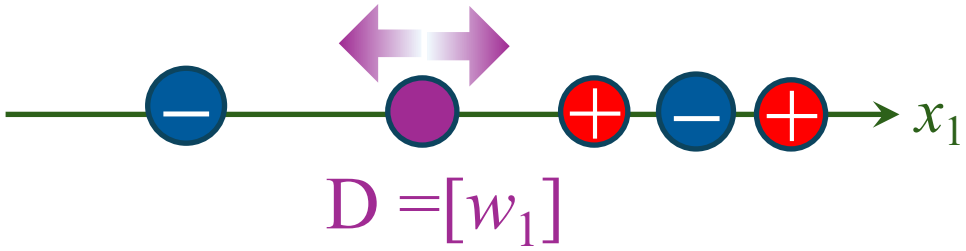


A **rope team** is a climbing technique where two or more climbers who are attached to a single climbing rope move simultaneously together along easy-angled terrain that does not require points of fixed climbing protection to be inserted along the route.

Learning the Discriminator: Multiple Observations

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$

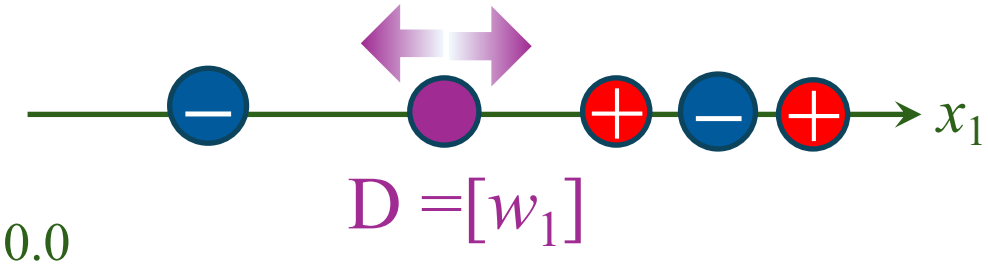


One step down based on which ones?

Learning the Discriminator: Multiple Observations

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$



One step down per pair of [one positive, one negative]:

$$w^{(1)}{}_1 \rightarrow w^{(0)}{}_1 - \eta \times \frac{d\, e(w^{(0)}{}_1 : (a^{(0)}{}_1, b^{(0)}{}_1))}{d\, w_1}$$

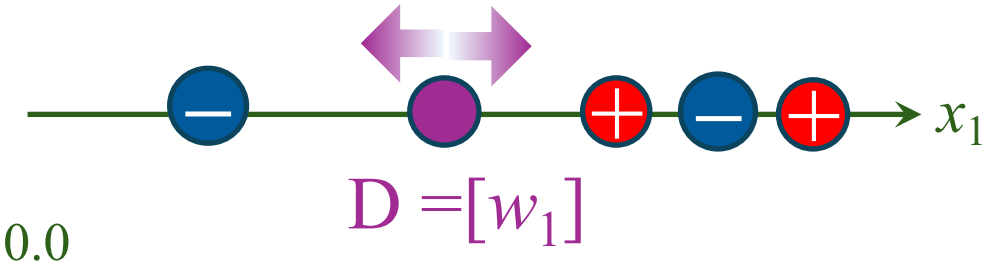
$$w^{(2)}{}_1 \rightarrow w^{(1)}{}_1 - \eta \times \frac{d\, e(w^{(1)}{}_1 : (a^{(1)}{}_1, b^{(1)}{}_1))}{d\, w_1}$$

$$w^{(3)}{}_1 \rightarrow w^{(2)}{}_1 - \eta \times \frac{d\, e(w^{(2)}{}_1 : (a^{(2)}{}_1, b^{(2)}{}_1))}{d\, w_1}$$

Learning the Discriminator: Multiple Observations

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$



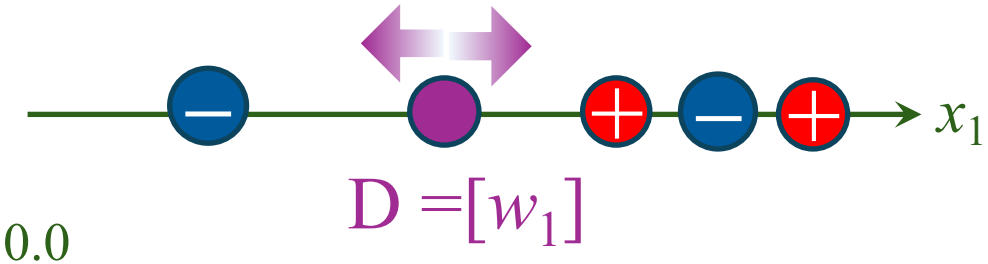
One **average** step down of **all** pairs :

$$w^{(1)}{}_1 \rightarrow w^{(0)}{}_1 - \eta \times \frac{1}{n} \sum_{a_1 \in \mathbf{O}^+,\, b_1 \in \mathbf{O}^-} \frac{d\, e(w^{(0)}{}_1 : (a_1, b_1))}{d\, w_1}$$

Learning the Discriminator: Multiple Observations

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$



One **average** step down of pairs in subsets, aka **batches**:

$$w^{(1)}{}_1 \rightarrow w^{(0)}{}_1 - \eta \times \frac{1}{batch\ size} \sum_{a_1, b_1 \in batch\ 1} \frac{d\ e(w^{(0)}{}_1 : (a_1, b_1))}{d\ w_1}$$

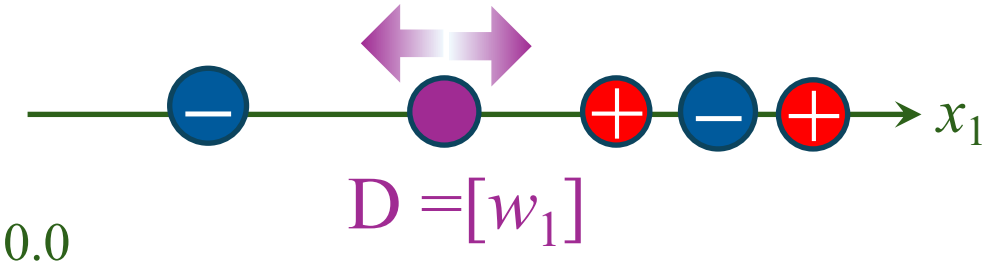
$$w^{(2)}{}_1 \rightarrow w^{(1)}{}_1 - \eta \times \frac{1}{batch\ size} \sum_{a_1, b_1 \in batch\ 2} \frac{d\ e(w^{(1)}{}_1 : (a_1, b_1))}{d\ w_1}$$

$$w^{(3)}{}_1 \rightarrow w^{(2)}{}_1 - \eta \times \frac{1}{batch\ size} \sum_{a_1, b_1 \in batch\ 3} \frac{d\ e(w^{(2)}{}_1 : (a_1, b_1))}{d\ w_1}$$

Learning the Discriminator: Multiple Rounds: Epochs

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$



1st **full** consideration of all observations: **1 epoch**
2nd **full re-**consideration of all observations: **2 epochs**
3rd **full re-**consideration of all observations: **3 epochs**
...
N-th **full re-**consideration of all observations: **N epochs**

Learning the Discriminator: Training Error Trend

Error Function $e(w)$ = Loss Function $\mathcal{L}(w)$ = Objective Function

Step Down = Gradient Descend = A Technique to Optimize Functions

